# Unifying Theory of GLMs

Rebecca C. Steorts (slide adaption from Maria Tacket) and material from Chapter 5 of Roback and Legler text.

# Agenda

1. Review Poisson GLM (and any questions for upcoming quiz).
2. Reminder (homework due this week).
3. Exercises for exponential families.
4. Continue with GLM theory.

# Computing set up

```r
library(tidyverse)
library(tidymodels)
library(knitr)
library(viridis)

knitr::opts_chunk$set(fig.width = 8,
                      fig.asp = 0.618,
                      fig.retina = 3,
                      dpt = 300,
                      out.width = "70%",
                      fig.align = "center")

ggplot2::theme_set(ggplot2::theme_bw(base_size = 16))

colors <- tibble::tibble(green = "#B5BA72")
```

# Topics

- Identify the components common to all generalized linear models
- Find the canonical link based on the distribution of the response variable
- Properties of GLMs

Notes based on Chapter 5 Roback and Legler (2021) unless noted otherwise.

# Unifying theory of GLMs

# Many models; one family

We have studied models for a variety of response variables

- Least squares (Normal)
- Logistic (Bernoulli, Binomial, Multinomial)
- Log-linear (Poisson, Negative Binomial)

These models are all examples of **generalized linear models**.

GLMs have a similar structure for their likelihoods, MLEs, variances, so we can use a generalized approach to find the model estimates and associated uncertainty.

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2. A linear combination of predictors, $\eta = \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$ (**systematic component**)

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2. A linear combination of predictors, $\eta = \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$ (**systematic component**)
3. A **link** function $g(\theta)$ that connects $\theta$ to $\eta$

# One-parameter exponential family form

Suppose a probability (mass or density) function has a parameter $\theta$. It is said to have a **one-parameter exponential family form** if

- The support (set of possible values) does not depend on $\theta$, and
- The probability function can be written in the following form

$$f(y;\theta) = e^{[a(y)b(\theta)+c(\theta)+d(y)]}$$

# Mean and variance

One-parameter exponential family form

$$f(y;\theta) = e^{[a(y)b(\theta)+c(\theta)+d(y)]}$$

Using this form:

$$E(Y) = -\frac{c'(\theta)}{b'(\theta)} \qquad Var(Y) = \frac{b''(\theta)c'(\theta) - c''(\theta)b'(\theta)}{[b'(\theta)]^3}$$

# Poisson in one-parameter exponential family form

$$P(Y = y) = \frac{e^{-\lambda}\lambda^y}{y!} \qquad y = 0, 1, 2, \ldots, \infty$$

$$\begin{aligned} P(Y = y) &= e^{-\lambda} e^{y \log(\lambda)} e^{-\log(y!)} \\ &= e^{y \log(\lambda) - \lambda - \log(y!)} \end{aligned}$$

Recall the form: $f(y; \theta) = e^{[a(y)b(\theta) + c(\theta) + d(y)]}$, where the parameter $\theta = \lambda$ for the Poisson distribution

- $a(y) = y$
- $b(\lambda) = \log(\lambda)$
- $c(\lambda) = -\lambda$
- $d(y) = -\log(y!)$

# Poisson in exponential family form

- The support for the Poisson distribution is $y = 0, 1, 2, \ldots, \infty$. This does not depend on the parameter $\lambda$.
- The probability mass function can be written in the form $f(y; \theta) = e^{[a(y)b(\theta)+c(\theta)+d(y)]}$

**The Poisson distribution can be written in one-parameter exponential family form.**

# Canonical link

Suppose there is a response variable $Y$ from a distribution with parameter $\theta$ and a set of predictors that can be written as a linear combination $\eta = \beta_0 + \sum_{j=1}^{p} \beta_j x_j = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$

- A **link function**, $g()$, is a monotonic and differentiable function that connects $\theta$ to $\eta$
- When working with a member of the one-parameter exponential family, $b(\theta)$ is called the **canonical link**
- Most commonly used link function

# Canonical link for Poisson

Recall the exponential family form:

$$P(Y = y) = e^{y \log(\lambda) - \lambda - \log(y!)}$$

then the canonical link is $b(\lambda) = \log(\lambda)$

# GLM framework: Poisson response variable

1. Response variable with parameter $\theta$ whose probability function can be written in exponential family form

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$$\eta = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

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1. Response variable with parameter $\theta$ whose probability function can be written in exponential family form

$$P(Y = y) = e^{y \log(\lambda) - \lambda - \log(y!)}$$

2. A linear combination of predictors,

$$\eta = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

3. A function $g(\lambda)$ that connects $\lambda$ and $\eta$

$$\log(\lambda) = \eta = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

# Activity: Generalized linear models

For your group's distribution

- Write the pmf or pdf in one-parameter exponential form.
- Describe an example of a setting where this random variable may be used.
- Identify the canonical link function.

# Activity: Generalized linear models

**Distributions**

1. Exponential
2. Gamma (with fixed $r$)
3. Geometric
4. Binary

See BMLR - Section 3.6 for details on the distributions.

If your group finishes early, try completing the exercise for another distribution.

# Using the exponential family form

The one-parameter exponential family form is utilized for

- Calculating MLEs of coefficients (recall iteratively re-weighted least squares)
- Inference for coefficients
- Likelihood ratio and drop-in-deviance tests

The specific calculations are beyond the scope of this course. See Section 4.6 of Dunn, Smyth, et al. (2018) for more detail (available at Duke library).

# Exponential Distribution

Let Y = time spent waiting for the first event in a Poisson process with an average rate $\lambda$ events per time unit.

$$f(y, \lambda) = \lambda \exp\{-\lambda y\}.$$

# Exponential Distribution

a. Write the pmf or pdf in one-parameter exponential form and c. give the canonical link.

$$
\begin{aligned}
f(y, \lambda) &= \lambda \exp\{-\lambda y\} && (1)\\
&= \exp\{\log(\lambda \exp\{-\lambda y\})\} && (2)\\
&= \exp\{\log \lambda - \lambda y\} && (3)
\end{aligned}
$$

Recall the form: $f(y; \theta) = e^{[a(y)b(\theta)+c(\theta)+d(y)]}$, where the parameter $\theta = \lambda$ for the Exponential distribution.

- $a(y) = -y$
- $b(\lambda) = \lambda =$ canonical link
- $c(\lambda) = \log(\lambda)$
- $d(y) = 0$

# Exponential Distribution

b. Example: The exponential distribution can be used to model the number of miles traveled until encountering the first pothole on a North Carolina road.

# Gamma distribution (with fixed $r$)

$\mathrm{Y}$ = time spent waiting for the $r$th event in a Poisson process with an average rate of $\lambda$ events per unit of time.

$$f(y, \lambda) = \frac{\lambda^r}{\Gamma(r)} y^{r-1} \exp\{-\lambda y\}.$$

# Gamma distribution (with fixed $r$)

a. Write the pmf or pdf in one-parameter exponential form and c. give the canonical link.

$$
\begin{aligned}
f(y,\lambda) &= \frac{\lambda^r}{\Gamma(r)} y^{r-1} \exp\{-\lambda y\} && (4)\\
&= \exp\left[\log\left(\frac{\lambda^r}{\Gamma(r)} y^{r-1} \exp\{-\lambda y\}\right)\right] && (5)\\
&= \exp\left[r\log\lambda - \log(\Gamma(r)) + (r-1)\log y - \lambda y\right] && (6)\\
&\propto \alpha \exp\left[-\lambda y + r\log\lambda + (r-1)\log y\right] && (7)
\end{aligned}
$$

Recall the form: $f(y;\theta) = e^{[a(y)b(\theta)+c(\theta)+d(y)]}$, where the parameter $\theta = \lambda$ for the Gamma distribution.

- $a(y) = -y$
- $b(\lambda) = \lambda =$ canonical link
- $c(\lambda) = r\log(\lambda)$
- $d(y) = (r-1)\log y$

# Gamma distribution (with fixed $r$)

b. Example: The gamma distribution can be used to model the number of miles traveled until encountering 10 potholes on a North Carolina road.

# Geometric distribution

Y = number of failures before the first success in a Bernoulli process

$$f(y, p) = (1 - p)^y p.$$

# Geometric distribution

a. Write the pmf or pdf in one-parameter exponential form and c. give the canonical link.

$$
\begin{aligned}
f(y, \lambda) &= (1-p)^y p && (8)\\
&= \exp\{\log[(1-p)^y p]\} && (9)\\
&= \exp\{y \log(1-p) + \log(p)\} && (10)
\end{aligned}
$$

Recall the form: $f(y;\theta) = e^{[a(y)b(\theta)+c(\theta)+d(y)]}$, where the parameter $\theta = \log(1-p)$ for the Geometric distribution.

- $a(y) = y$
- $b(p) = \log(1-p) =$ canonical link
- $c(p) = \log(p)$
- $d(y) = 0$

# Geometric distribution

b. Example: A geometric distribution can be used to model the number of random people you call who decline before someone agrees to complete a survey.

# Binary distribution

$$f(y, p) = p^y (1 - p)^{1-y}$$

# Binary distribution

a. Write the pmf or pdf in one-parameter exponential form and c. give the canonical link.

$$
\begin{aligned}
f(y,p) &= p^y(1-p)^{1-y} && (11)\\
&= \exp\{\log[p^y(1-p)^{1-y}]\} && (12)\\
&= \exp\{y\log p + (1-y)\log(1-p)\} && (13)\\
&= \exp\{y\log(\frac{p}{1-p}) + \log(1-p)\} && (14)
\end{aligned}
$$

Recall the form: $f(y;\theta) = e^{[a(y)b(\theta)+c(\theta)+d(y)]}$, where the parameter $\theta = \log(\frac{p}{1-p})$ for the Binary distribution.

- $a(y) = y$
- $b(p) = \log(\frac{p}{1-p}) =$ canonical link
- $c(p) = \log(1-p)$
- $d(y) = 0$

# References


Dunn, Peter K, Gordon K Smyth, et al. 2018. *Generalized Linear Models with Examples in r*. Vol. 53. Springer.

Nelder, John Ashworth, and Robert WM Wedderburn. 1972. "Generalized Linear Models." *Journal of the Royal Statistical Society Series A: Statistics in Society* 135 (3): 370–84.

Roback, Paul, and Julie Legler. 2021. *Beyond multiple linear regression: applied generalized linear models and multilevel models in R*. CRC Press.